

# Regime-Switching Models for EU Carbon Allowance Prices

## 1. Introduction

The European Union Emissions Trading System (EU ETS) is the world's largest carbon market. It operates as a cap-and-trade system where a fixed number of emissions allowances are issued each year and firms buy and sell these allowances to meet their compliance obligations. The price of an allowance, called a European Union Allowance (EUA), is set by supply and demand in that market. Since the market launched in 2005, EUA prices have ranged from near zero to over €100 per ton, driven by policy changes, energy market shocks, and macroeconomic events such as the 2008 financial crisis, the COVID-19 pandemic, and the 2022 energy crisis following Russia's invasion of Ukraine.

Carbon pricing systems like the EU ETS are important for meeting global climate targets, but only reduce emissions if prices stay high and stable to push firms toward cleaner alternatives. This makes understanding how these prices actually behave a question with direct consequences for climate policy and sustainability.

There is a lot of randomness in this system. EUA prices respond to fuel price movements, weather patterns affecting energy demand, and announcements of new climate regulations, which are all susceptible to randomness. EUA prices are also visibly inconsistent and cycle through distinct periods from slow drift to sharp crashes or rapid run-ups. This behavior makes EU ETS prices a good fit for stochastic modeling, and specifically for the models that I'm interested in as a data science major. My aim for this project is to model EUA spot prices using a Markov regime-switching framework using the tools of stochastic processes, specifically finite-state Markov chains and stationary distributions.

## 2. Questions of Interest

1. What are the long-run probabilities of each market regime? If the market can be in one of three states, normal, stressed, and crashed, the stationary distribution  $\pi = (\pi_1, \pi_2, \pi_3)$  of the regime-switching Markov chain gives the long-run fraction of time spent in each state. Calculating  $\pi$  from the estimated transition matrix tells us how often the market is in a distressed state over a multi-year horizon.
2. How long does the market stay in each regime once it enters? The expected time spent in regime  $i$  before the chain transitions out is  $1/(1 - p_{ii})$ , where  $p_{ii}$  is the diagonal entry of  $P$ . How many days on average pass before the chain leaves state  $i$  (time hitting question)?
3. How sensitive is the stationary distribution to changes in the transition probabilities? Policy interventions are designed to reduce the frequency and severity of price crashes. Do changes in the estimated transition probabilities across the EU ETS trading phases correspond to meaningful changes in  $\pi$ ?
4. Can the hidden regime be identified from observed prices? Because the regime is latent, can the Markov chain's state at each point in time be estimated from observed price data? This motivates the use of a hidden Markov model decoding application.

### 3. Description of the Random Dynamics

The state of the system at time  $t$  is the daily EUA spot price  $X_t$ . What evolves over time is both the price level and the volatility of price changes, which is what makes a single fixed model inadequate. The key modeling assumption is that there is a hidden variable  $S_t$  taking values in  $\{1, 2, 3\}$ , representing three market regimes: (1) a normal regime characterized by slow mean-reverting drift and low volatility, (2) a stressed regime with elevated volatility and faster price movement, and (3) a crashed regime with low prices and low activity. The regime  $S_t$  is not observed directly, only  $X_t$  is observed.

There are two instances of random dynamics. First,  $S_t$  itself evolves randomly. Each day, the market either stays in its current regime or transitions to another, with probabilities given by a  $3 \times 3$  transition matrix  $P$ . This transition is the Markov chain at the center of the model, as the probability of tomorrow's regime depends only on today's regime, not on the full history of prices. The entries of  $P$  represent how likely the market is to remain stressed or return to normal after a shock, or how rarely the crashed regime is entered from the normal regime directly. Second, conditional on the regime, the price  $X_t$  evolves according to a regime-specific process with its own mean and variance. In regime 1 (normal), changes in log price are small and approximately Gaussian. In regime 2 (stressed), the variance is larger and prices move more erratically. In regime 3 (crashed), the price sits near a low level with little movement. Rather than fitting one set of parameters to the full price history, each regime captures a qualitatively different market environment.

The deterministic structure is the number of regimes, the form of the within-regime dynamics, and the constraint that transitions follow the Markov property. What is random is which regime the market is in at any given time, and the price fluctuations within that regime. Because the regime is never directly observed, the model must infer it from the sequence of prices alone. This is the hidden Markov model structure where a latent Markov chain drives an observable process.

The project will fit this model to historical EUA spot price data from Phase II onward (2008 to present) using maximum likelihood estimation with the Baum-Welch (EM) algorithm, a standard method for hidden Markov models. The estimated transition matrix will then be used to compute the stationary distribution, expected regime durations, and a decoded sequence of most-likely regimes across the sample period.

Daily EUA spot price data are publicly available from the International Carbon Action Partnership (ICAP) Allowance Price Explorer at [ICAP Carbon Action](#) and from the European Environment Agency data hub at [EEA Europe EU Datahub](#). The analysis will be implemented in Python using the `hmmlearn` library for hidden Markov model estimation. All code and processed data will be made available in a public GitHub repository.

### 4. References

Frontiers in Environmental Science, "Carbon market volatility analysis based on structural breaks: Evidence from EU-ETS and China" (2022). <https://doi.org/10.3389/fenvs.2022.973855>

Bisiotis et al., "Forecasting Carbon Prices: A Literature Review," *Journal of Forecasting* (2026). <https://doi.org/10.1002/for.70054>